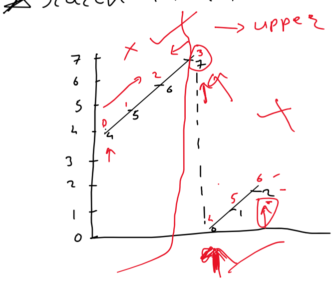
Lec29

25 March 2026 10:48 AM

Search in rotated sorted Array



→ upper

[4,5,6,7,0,1,2]

↓ Lo ↓ hi

- item = 0

Lo = 0

Hi = 6

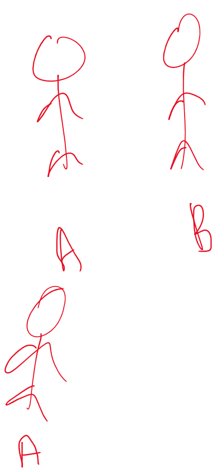
$\frac{0+6}{2} = 3$

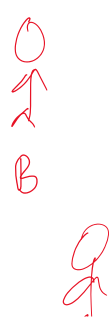
upper

↓

arr[lo] < arr[mid]

```
while (lo<=hi){
    mid = (lo+hi)/2
    if(arr[mid] == item){
        return mid
    }
    else if(arr[lo] <=arr[mid])){
        if (arr[lo] <= item && item < arr[mid])){
            hi = mid-1
        }else{
            lo = mid+1
        }
    }else{
        if(item > arr[mid] && item <= arr[hi]){
            lo = mid+1
        }else{
            hi = mid-1
        }
    }
}
```





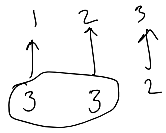
A

B

cand1 = 1
candi2 = 2

count1 = ~~0 1 2~~ 3
count2 = ~~0 1 2~~ 3

[2, 2, 3, 1, 3, 2, 1, 1]



n/3

num	candi1	candi2	count1	count2
2	2	0	1	0
2	2	0	2	0
3	2	3	1	0
1	2	1	0	0
3	3	0	1	0
2	3	2	0	0
1	1	2	1	0
1	1	2	2	0

```
List<Integer> list = new A

if (count1 > n/3){
    list.add(candi1
}
if (count2 > n/3){
    list.add(cand
}
return list.
```

A

B

X —

rrayList<>()

i2